

Exact-Cost Development Screening and Risk-Constrained Evaluation of Safety-Monitor Cascades

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Abstract

Selective safety-monitor cascades are easily overstated when policies are compared at unequal realized cost, when evaluation ranks determine acquisition, or when false-positive-rate constraints are treated as point-estimate filters rather than risks requiring independent control. We re-evaluate a cost-aware safety-monitor router under a frozen protocol that requires deployable online thresholds, repeated grouped splits, heterogeneous latency, exact-cost equivalence, stronger baselines, and independent risk control before any superiority claim. The permitted development view contains 1,687 examples. Three signed-value model families are compared with threshold distance, current-error prediction, deterministic random acquisition, direct fusion, and never/always-acquire endpoints. No signed-value family has a jointly feasible development operating point, and each family passes only one of sixteen primary pairwise cost-equivalence comparisons. Consequently no router family is selected. Controlled single-T4 timing shows that optional-monitor execution dominates runtime and has a substantial upper tail. Fresh calibration and fresh source- and time-separated multi-rater confirmatory data were not available, so no joint-risk certificate or router-superiority claim is reported. The resulting contribution is an evaluation and measurement study showing how apparent routing gains can disappear under stricter cost, risk, and deployment-valid comparison.

Keywords: safety monitoring; selective acquisition; exact cost; risk control; decision value; latency measurement

1 Introduction

Runtime safety systems often combine inexpensive monitors with a stronger but substantially slower optional monitor. A selective router is useful only if the examples chosen for acquisition produce enough downstream decision benefit to justify both the added latency and the associated risk. Evaluating such a router therefore requires more than a recall-cost curve.

An earlier development frontier in this project appeared favorable under a cost-ceiling comparison and an empirical FPR screen. That frontier is not treated as evidence for router superiority here. The comparison did not place every selective policy at the same realized absolute cost, did not define a single jointly feasible random policy, did not provide selection-valid joint FPR and cost control, and used ranked evaluation outputs rather than a frozen deployable online acquisition threshold. The present analysis therefore replaces that claim with a stricter evaluation design.

The objective is deliberately conditional. A router may support a superiority claim only if it survives the development comparison, can be frozen on independent calibration data, passes joint FPR and mean-cost control, and is then confirmed once on fresh data. If the development evidence already fails the exact-cost comparison, the appropriate result is a no-go for the router claim rather than additional tuning on the same data.

2 Evaluation Design

2.1 Policy-specific signed decision value

Let H denote all information available before acquisition, Z the optional-monitor output, Y the target label, $\pi_{\text{base}}(H)$ the frozen base policy, and $\pi_{\text{aug}}(H, Z)$ the policy available after acquisition. For loss L , realized acquisition value is

$$V_{\pi} = L(Y, \pi_{\text{base}}(H)) - L(Y, \pi_{\text{aug}}(H, Z)).$$

The corresponding pre-acquisition target is

$$v_{\pi}(h) = \mathbb{E}[V_{\pi} \mid H = h].$$

This quantity is policy-specific and signed. Under zero-one loss, $V_{\pi} \in \{-1, 0, 1\}$: acquisition can correct the base decision, leave it unchanged, or make it worse. It is therefore incorrect to replace the fitted-policy estimand with a generic nonnegative Bayes value.

With heterogeneous incremental latency $c(h)$, the online cost-aware ranking score is

$$s(h) = \frac{\hat{v}_{\pi}(h)}{\max\{\hat{c}(h), 1 \text{ ms}\}}.$$

Only pre-acquisition variables enter $\hat{v}_{\pi}$ and $\hat{c}$. Optional-monitor outputs and post-acquisition latency are excluded from router features.

2.2 Development data and grouped resampling

Development uses only the authorized legacy development view: `policy_train`, `policy_selection`, and `calibration`. Together these contain 1,687 examples, including 291 positives and 1,396 negatives.

The protected legacy `final_test` and `held_out_shift` partitions remain unopened in the v2 analysis. They are not substitutes for fresh confirmation.

Repeated development analysis uses five `StratifiedGroupKFold` seeds: 1729, 2718, 3141, 5772, and 8111, with five outer folds and four inner folds. Effective groups remain intact. Hyperparameters are chosen only inside the current inner training partition.

For every held-out outer fold, acquisition thresholds are calibrated using only the other four out-of-fold partitions for the same seed. Thus the evaluated examples do not determine their own acquisition rank or threshold.

2.3 Policies and comparators

The proposed router is evaluated using three signed-value regressors:

- Ridge;
- HistGradientBoostingRegressor;
- RandomForestRegressor.

The required selective comparators are:

- distance to the frozen base decision threshold;

- current-error prediction, which estimates whether the frozen base policy is wrong;
- deterministic SHA-256 random acquisition using the five prespecified policy seeds.

Direct full-information fusion is retained as the full-cost endpoint rather than a selective same-cost superiority target. Never-acquire and always-acquire policies provide the remaining reference endpoints.

2.4 Development exact-cost screening

The protocol defines primary cost as total policy latency from input preparation through the final decision. Selective policies must be compared at the same absolute millisecond budget. Adjacent streaming thresholds can be mixed with deterministic hash-based boundary randomization so that expected calibration cost matches the target without test-set ranking.

For the present development screening, fresh calibration data were not available. The six operating fractions 0.10, 0.20, 0.30, 0.50, 0.75, 1.00 were therefore anchored only to the measured development component totals. The mean base component total was 43.78 ms, the always-acquire component total was 3,746.4 ms, and the measured increment was 3,702.6 ms. The resulting development targets were 414.0, 784.3, 1,154.6, 1,895.1, 2,820.8, 3,746.4 ms.

These targets are development preselection quantities, not final deployment budgets. Final absolute budgets must be constructed from the independent fresh-calibration always-acquire increment.

At each development target, the primary cost comparison is a paired two-one-sided group-bootstrap equivalence test. The equivalence margin is

$$\delta_B = \max(1 \text{ ms}, 0.01B),$$

where B is the absolute target. Recall comparison is admissible only after cost equivalence passes. The development stability rule also requires the primary recall difference to retain the same sign in at least four of five seeds.

2.5 Risk-control claim boundary

The formal deployment protocol separates threshold construction from risk certification. Fresh calibration is split by effective group into a 30% optimization subset and a 70% risk-testing subset. Candidate policies are Pareto filtered before Learn Then Test joint control [2] of

$$\text{FPR} \leq 0.05$$

and

$$\mathbb{E}[C_{\text{E2E}}] \leq B.$$

FPR control uses a one-sided exact-binomial test [1]. The candidate joint p -value is the maximum of the two constraint p -values, with Bonferroni control across the Pareto-filtered candidate set. The bounded mean-cost test uses Hoeffding–Bentkus concentration [3, 4]. A point estimate alone cannot certify a router.

Because qualifying fresh calibration data were unavailable, this formal joint-risk step was not executed. Development FPR values are reported only as descriptive evidence.

A router-superiority claim would additionally require one sealed fresh confirmatory evaluation with source and time separation, deduplication, stronger multi-rater labels, a one-sided 95% FPR upper bound at most 0.05, matched-cost equivalence against each primary comparator, and Holm-adjusted paired recall improvement [6]. Those confirmatory conditions are not evaluated here.

Table 1: Development signed-value family selection. “Feasible budgets” refers only to the development screening rule, not to fresh joint-risk certification.

Family	Feasible budgets	Cost-equivalence passes	Router + cost predictor (ms)
Ridge	0	1/16	0.219
HistGradientBoosting	0	1/16	3.277
RandomForest	0	1/16	40.95

2.6 Controlled latency measurement

Latency was measured on a single NVIDIA Tesla T4 at batch size one with 20 warm-up requests. Accelerator stages were synchronized around timing boundaries and raw per-example timings were retained. Tail confidence intervals use 2,000 effective-group bootstrap resamples [5].

The measured stages include input preparation, base monitoring, router-feature construction, router inference, optional-monitor execution, and final fusion. Component totals reported below are sums of directly measured non-overlapping stages. They are not relabeled as direct wall-clock end-to-end measurements for a selected online router because no router survives development selection.

3 Results

3.1 Development selection is a no-go

Table 1 summarizes the exact-cost development selection. None of the three signed-value families has a jointly feasible development budget. Each family passes only one of sixteen primary pairwise cost-equivalence comparisons.

The development result is therefore not a choice among several successful routers. Under the frozen comparison rule there is no eligible signed-value family to freeze. Additional model or threshold tuning on these same development folds would change the scientific question after observing the outcome.

Figure 1 shows the raw mean recall differences. Several operating points have positive signed-value-minus-comparator recall differences, but those differences cannot be interpreted as same-cost superiority when the corresponding equivalence test fails. Square outlines identify points that pass the cost- equivalence test.

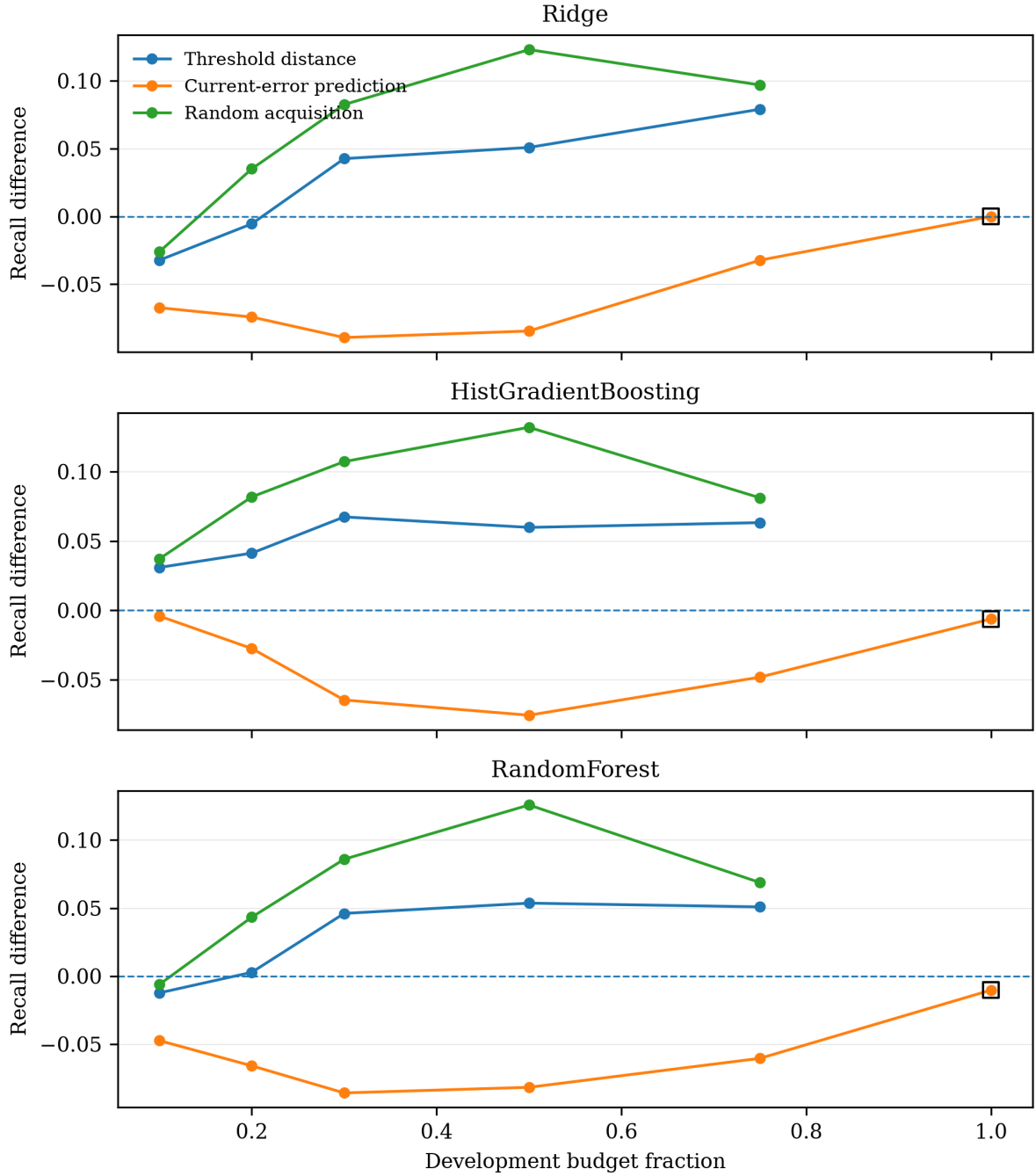


Figure 1: Mean development recall difference between each signed-value family and the primary selective comparators. Positive values alone do not establish improvement; recall comparison is admissible only after paired cost equivalence. Square outlines denote cost-equivalence passes.

Figure 2 makes the principal failure mode explicit: the cost-equivalence pass rate is 1/16 for every signed-value family.

Table 2: Development endpoint metrics averaged across five fold seeds. Costs are measured component totals. FPR values are descriptive development measurements, not finite-sample risk certificates.

Endpoint	Family	Mean cost (ms)	Recall	Mean FPR	Maximum seed FPR
never acquire	base policy	43.78	0.2179	0.0504	0.0516
always acquire	augmented policy	3,746.4	0.8570	0.0470	0.0494
direct fusion	HistGradientBoosting	3,746.5	0.8667	0.0490	0.0509
direct fusion	Logistic regression	3,743.9	0.8598	0.0473	0.0494
direct fusion	RandomForest	3,781.9	0.8577	0.0491	0.0509

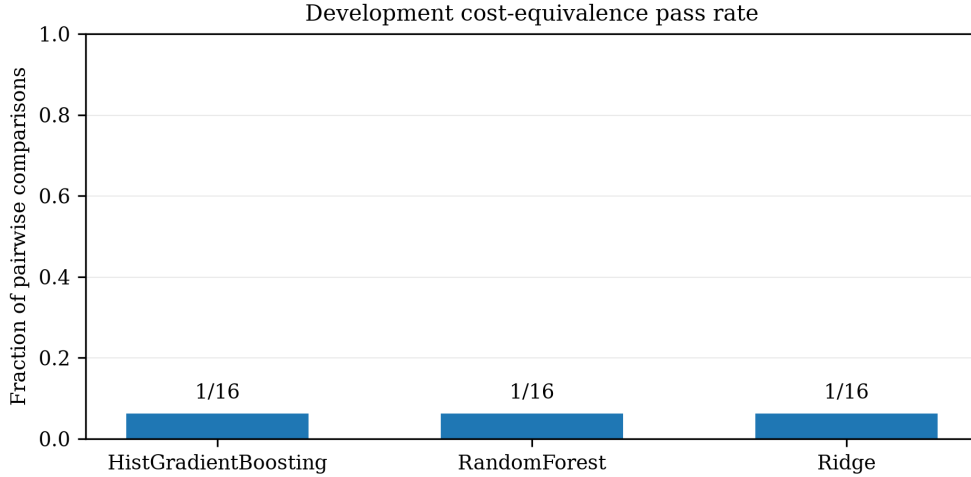


Figure 2: Fraction of primary development comparisons passing the paired cost-equivalence test. All three signed-value model families pass one of sixteen comparisons.

3.2 Reference endpoints

Table 2 reports the reference endpoints averaged across the five fold seeds. Never acquisition has mean recall 0.2179 and mean FPR 0.0504. Always acquisition increases mean recall to 0.8570 at an average component total of 3,746.4 ms. Direct-fusion models operate at approximately the full acquisition cost and are therefore reported as full-cost endpoints rather than selective same-cost comparators.

The base endpoint itself has maximum seed FPR 0.0516, slightly above 0.05. Some direct-fusion models also cross 0.05 in at least one seed. These observations reinforce the distinction between development point estimates and formal risk control.

3.3 Latency heterogeneity and upper-tail cost

The optional monitor dominates runtime. Its measured mean is 3,702.8 ms, median 3,423.3 ms, p95 5,911.7 ms, p99 7,456.4 ms, and maximum 22,809.5 ms. By comparison, the mean base-monitor stage is 40.71 ms and the complete-text embedding stage averages 77.27 ms.

Router inference is comparatively inexpensive for Ridge (0.219 ms) and HistGradientBoosting (3.277 ms), while the RandomForest signed-value router is materially heavier at 40.95 ms. These differences are small relative to optional-monitor execution but remain part of the online acquisition cost.

Table 3: Controlled single-T4 latency summary in milliseconds. Tail confidence intervals use 2,000 effective-group bootstrap resamples.

Stage	<i>n</i>	Mean	Median	p95	95% CI p95	p99	95% CI p99	Maximum
Base monitors	1687	40.71	36.08	58.45	[58.18, 58.84]	61.12	[60.84, 62.40]	75.22
Embedding	1687	77.27	66.98	162.7	[158.7, 166.9]	209.1	[202.5, 240.3]	440.4
Optional monitor	1687	3,702.8	3,423.3	5,911.7	[5,655.7, 5,960.9]	7,456.4	[6,856.7, 8,025.7]	22,809.5
Ridge router inference	8435	0.219	0.214	0.261	[0.259, 0.263]	0.340	[0.320, 0.361]	0.554
HGBR router inference	8435	3.277	3.111	3.932	[3.911, 3.954]	4.516	[4.402, 4.760]	11.69
RF router inference	8435	40.95	40.59	44.25	[43.98, 44.51]	48.40	[47.54, 49.64]	82.02
Base component total	8435	43.78	39.21	61.60	[61.23, 62.02]	64.35	[63.92, 65.61]	78.64
Always-acquire component total	8435	3,746.4	3,462.7	5,975.6	[5,716.8, 6,023.1]	7,522.4	[6,936.5, 8,104.6]	22,887.9

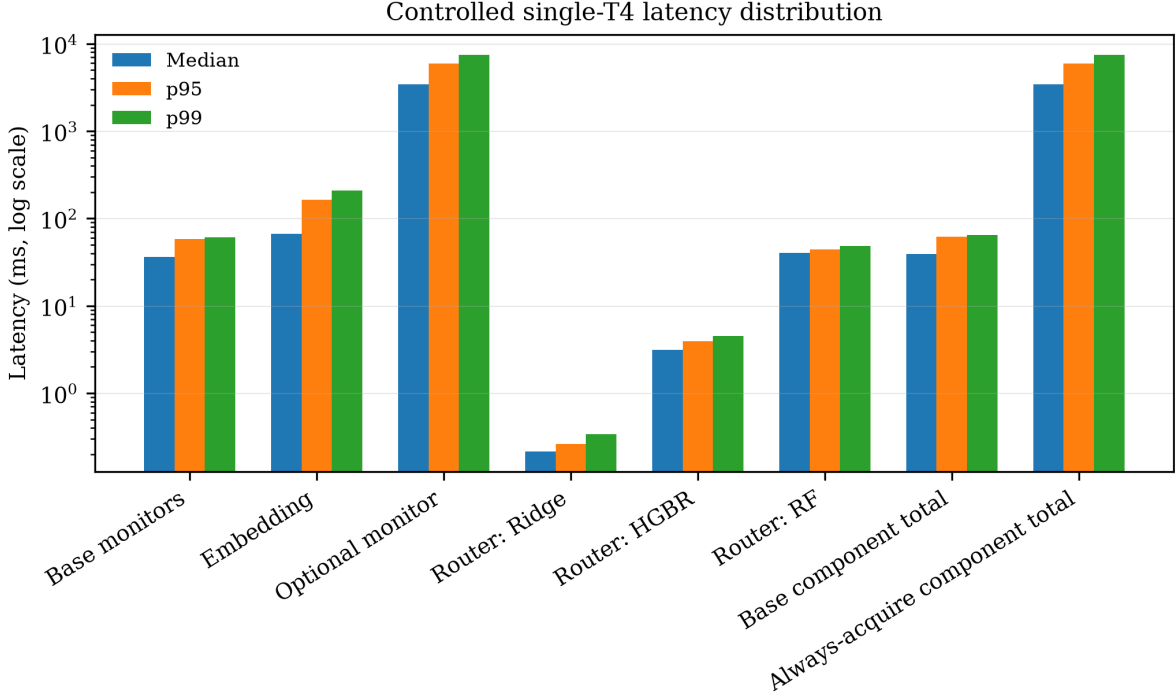


Figure 3: Controlled single-T4 latency distributions. Bars show median, p95, and p99 latency. The logarithmic scale exposes the separation between router inference and optional-monitor execution. Component totals are sums of directly timed non-overlapping stages.

No reported stage timed out in the controlled run. The absence of timeouts does not make the latency distribution homogeneous: optional-monitor p99 remains substantially above its median.

4 Discussion

The central result is methodological rather than a new routing claim. Predictive performance of a routing score is not sufficient to establish a better cascade. The policy must also survive comparison at matched realized cost, under a threshold that can actually be used online, with repeated grouped uncertainty and a risk-control procedure that is independent of the evaluation set.

This distinction changes the interpretation of the earlier development frontier. The historical frontier showed that ranked development examples could produce favorable local recall under a cost ceiling. Under the stricter protocol, that evidence is not valid for router superiority because the comparator costs were not matched by the present exact-cost criterion, random acquisition was not

represented by one jointly feasible frozen policy, FPR was screened descriptively, and evaluation ranks participated in the routing rule. The historical result remains useful as an example of why an apparently favorable frontier can disappear under a more deployment-valid comparison.

The signed-value correction is also substantive. Acquisition value belongs to the pair of fitted policies. If the optional monitor can change a correct base decision into an incorrect augmented decision, then value is negative for that example. A nonnegative intrinsic “monitor value” therefore does not describe the deployed cascade.

The development no-go should not be interpreted as evidence that selective acquisition is impossible in general. It establishes the narrower and more defensible statement that the tested signed-value families do not provide a selectable policy under this frozen development comparison. Any renewed router study would require new data and a newly justified protocol rather than post-hoc adjustment to these development outcomes.

5 Limitations

The study intentionally leaves several claim gates unresolved.

First, final absolute budgets require fresh calibration. The development analysis therefore uses measured component-cost anchors only for preselection. These anchors cannot be reported as final deployment exact-cost certificates.

Second, no signed-value family survives development selection. Consequently there is no scientifically justified selected router for a direct policy-level wall-clock end-to-end benchmark. The latency evidence reported here consists of controlled per-stage measurements and component totals, with full tail summaries.

Third, independent fresh calibration data were unavailable, so the joint FPR-and-cost Learn Then Test certificate was not executed.

Fourth, fresh confirmatory data satisfying source separation, time separation, deduplication, and the prespecified multi-rater labeling requirements were unavailable. No confirmatory FPR upper bound, Holm-adjusted recall comparison, or one-shot router-superiority claim is therefore reported.

Finally, the existing protected legacy final and shift partitions remain sealed in v2. Their reuse would not supply the fresh evidence required by the protocol.

6 Conclusion

A stricter evaluation of the proposed safety-monitor router does not support a routing-performance claim. Across five repeated grouped seeds and three signed-value model families, no family has a jointly feasible development operating point, and each family passes only one of sixteen primary pairwise cost-equivalence tests. Positive raw recall differences at some operating points therefore cannot be converted into same-cost superiority claims.

The supported contribution is an evaluation and measurement methodology for safety-monitor cascades: define policy-specific signed value, use only pre-acquisition routing information, compare selective policies at the same realized cost, require cost equivalence before recall comparison, separate optimization from risk certification, measure heterogeneous and upper-tail latency, and reserve superiority claims for fresh confirmatory evidence. Under those requirements, the present router remains a no-go.

References

- [1] C. J. Clopper and E. S. Pearson, “The use of confidence or fiducial limits illustrated in the case of the binomial,” *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934.
- [2] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei, “Learn then Test: Calibrating Predictive Algorithms to Achieve Risk Control,” *The Annals of Applied Statistics*, vol. 19, no. 2, pp. 1641–1662, 2025.
- [3] W. Hoeffding, “Probability inequalities for sums of bounded random variables,” *Journal of the American Statistical Association*, vol. 58, no. 301, pp. 13–30, 1963.
- [4] V. Bentkus, “On Hoeffding’s inequalities,” *The Annals of Probability*, vol. 32, no. 2, pp. 1650–1673, 2004.
- [5] B. Efron, “Bootstrap methods: Another look at the jackknife,” *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.
- [6] S. Holm, “A simple sequentially rejective multiple test procedure,” *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.